

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Design and Develop Model (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO4	Assessment:	Assessment 1: Design and Develop Model
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
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Section / Batch:	AIML (2024)	Date:	27.08.2026

Code of Conduct

I Ch. Durga Prasad (192425305) that this submission is my original work and that I have adhered to the guidelines specified specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: CH. Durga Prasad

QUESTIONS: 5

Total Marks: 50

Annexure A

Design and Develop Model

1. Transfer Learning for Image Classification

Design and develop an image classification model using Transfer Learning with DenseNet. Use a pretrained DenseNet model as a feature extractor, add a custom classification layer, and evaluate the model's performance on a suitable dataset.

2. Fine-Tuning a Pre-Trained Model

Develop an image classification system using a pre-trained deep learning model and apply fine-tuning. Compare the performance of the model before and after fine-tuning and analyze the impact of unfreezing different layers.

3. Feature Extraction-Based Model

Design and develop a deep learning model using a pre-trained CNN as a feature extractor for a realworld classification problem. Extract deep features, train a custom classifier, and evaluate the effectiveness of transfer learning.

4. Medical Image Classification

Design and develop a transfer learning-based model for classifying medical images, such as pneumonia detection, skin disease classification, or brain tumor detection. Use a pre-trained model and evaluate its accuracy and other performance metrics.

5. RNN for Text Classification

Design and develop a Recurrent Neural Network (RNN) model for text classification. Use an appropriate dataset to classify text into different categories and analyze how sequential information is processed by the network.

6. Sentiment Analysis Using LSTM

Develop a sentiment analysis system using Long Short-Term Memory (LSTM). Classify user reviews or social media posts as positive, negative, or neutral and compare the results with a traditional RNN model.

7. Bidirectional RNN for Sequence Classification

Design and develop a Bidirectional RNN (BiRNN) model for sequence classification. Compare its performance with a standard RNN and explain how processing information in both forward and backward directions improves the model.

8. Next-Word Prediction Using LSTM

Develop an LSTM-based next-word prediction system. Train the model on a suitable text corpus and demonstrate how the model learns sequential patterns and predicts the most probable next word.

9. Encoder–Decoder Sequence-to-Sequence Model

Design and develop an Encoder–Decoder-based Sequence-to-Sequence model for an application such as text summarization, machine translation, or question-answer generation. Explain the role of the encoder and decoder in processing sequences.

10. Comparative Deep Learning Model

Design and develop multiple deep learning models for the same problem using Transfer Learning, RNN, BiRNN, or LSTM techniques. Compare their architectures and performance using appropriate evaluation metrics, and justify which model is most suitable for the selected application.

QUESTION 1 – TRANSFER LEARNING FOR IMAGE CLASSIFICATION

Problem

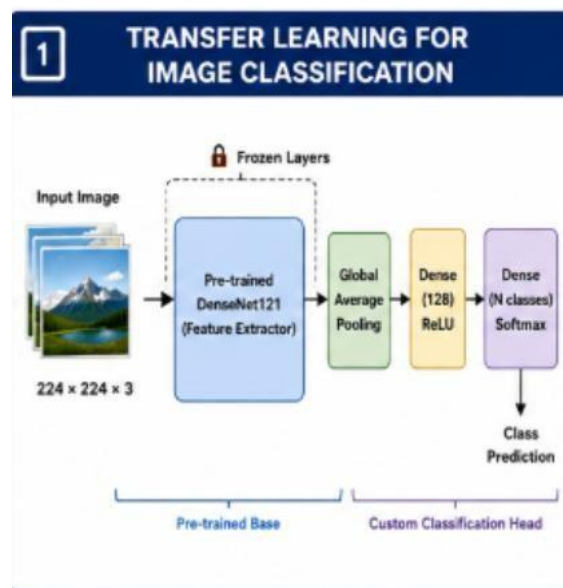
Design and develop an image classification model using Transfer Learning with DenseNet. Use a pre-trained DenseNet model as a feature extractor, add a custom classification layer, and evaluate the model.

Answer

Transfer Learning uses a model that has already been trained on a large dataset and adapts it to a new classification problem.

For this problem, **DenseNet121** can be used as the pre-trained model. A suitable dataset such as **CIFAR-10** can be used.

Architecture



Steps

1. Load the image dataset.
2. Resize images according to DenseNet input requirements.
3. Normalize pixel values.
4. Load pre-trained DenseNet121 without its original classifier.
5. Freeze the DenseNet layers.
6. Add Global Average Pooling.

7. Add a Dense classification layer.
8. Use Softmax for multi-class classification.
9. Train only the newly added layers.
10. Evaluate the model.

Example Code

```
import tensorflow as tf from tensorflow.keras.applications import
DenseNet121 from tensorflow.keras.layers import
GlobalAveragePooling2D, Dense from tensorflow.keras.models
import Model

base = DenseNet121(weights="imagenet",
                    include_top=False,
                    input_shape=(224,224,3))

base.trainable = False

x = GlobalAveragePooling2D()(base.output)
x = Dense(128, activation="relu")(x) output
= Dense(10, activation="softmax")(x)

model = Model(base.input, output)

model.compile(optimizer="adam",
              loss="sparse_categorical_crossentropy",          metrics=["accuracy"])

model.summary()
```

Evaluation

The model can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Conclusion

Transfer learning reduces training time and allows a pre-trained DenseNet to extract useful visual features. Freezing the base network also reduces the number of parameters that need to be trained.

QUESTION 2 – FINE-TUNING A PRE-TRAINED MODEL

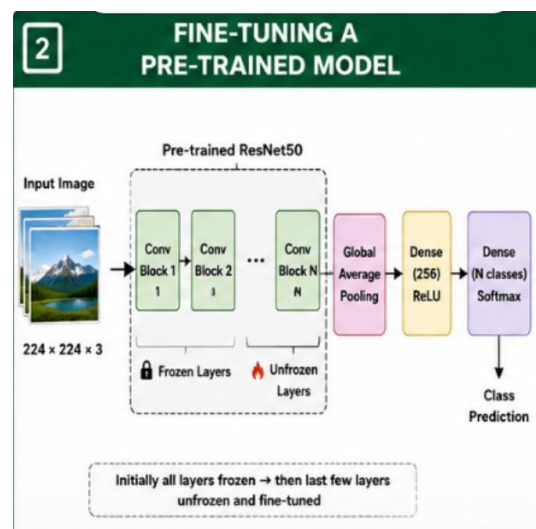
Problem

Develop an image classification system using a pre-trained deep learning model and apply fine-tuning. Compare performance before and after fine-tuning and analyze the effect of unfreezing different layers.

Answer

Fine-tuning is an extension of transfer learning where some layers of the pre-trained model are **unfrozen and trained on the new dataset**.

A pre-trained **ResNet50** can be used.



Stage 1 –

Architecture

Feature Extraction

Initially, all ResNet50 layers are frozen.

ResNet50 → Frozen



Custom Classifier



Prediction

Only the newly added classifier is trained.

Stage 2 – Fine-Tuning

The last few layers are unfrozen.

ResNet50

Early Layers → Frozen

Middle Layers → Frozen

Last Layers → Trainable



Custom Classifier

A small learning rate such as **0.0001** is used to avoid destroying previously learned features.

Comparison

Feature	Before Fine-Tuning	After Fine-Tuning
Base layers	Frozen	Some unfrozen
Training time	Lower	Higher
Adaptation	General features	Dataset-specific features
Accuracy	Good	Usually improved
Overfitting risk	Lower	Higher

Effect of Unfreezing Layers

- **Few layers unfrozen:** Faster and safer.
- **More layers unfrozen:** Better adaptation but greater overfitting risk.
- **All layers unfrozen:** Maximum adaptation but requires more data and computation.

Conclusion

Fine-tuning generally improves performance when the new dataset is sufficiently different from the original training data. A small learning rate and gradual unfreezing provide stable training.

QUESTION 3 – FEATURE EXTRACTION-BASED MODEL

Problem

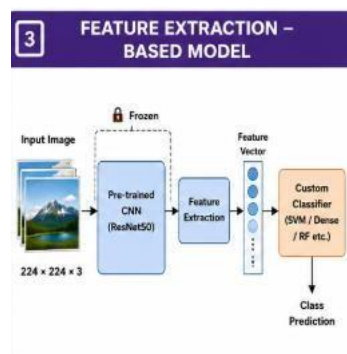
Design and develop a deep learning model using a pre-trained CNN as a feature extractor for a real-world classification problem.

Answer

A pre-trained CNN can be used to extract high-level image features. These extracted features can then be given to a custom machine learning classifier.

For example, **ResNet50** can be used for classifying different types of plant diseases.

Architecture



Process

1. Collect plant disease images.
2. Resize and normalize the images.
3. Load a pre-trained ResNet50.
4. Remove the original classification layer.
5. Freeze the CNN layers.
6. Pass images through the CNN.
7. Extract deep feature vectors.
8. Train a classifier using the extracted features.
9. Evaluate the classifier.

Why Feature Extraction?

Early CNN layers learn:

Edges → Textures → Shapes

Deep layers learn more meaningful patterns:

Shapes → Disease Patterns → Disease Features

Advantages

- Requires less training time.
- Uses previously learned visual features.
- Works well with limited datasets.
- Reduces computational requirements.
- Avoids training the complete CNN from scratch.

Evaluation

Suitable metrics include:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Conclusion

Feature extraction using a pre-trained CNN is an effective approach for real-world classification because powerful visual features can be reused without training a complete deep learning model from scratch.

QUESTION 4 – MEDICAL IMAGE CLASSIFICATION

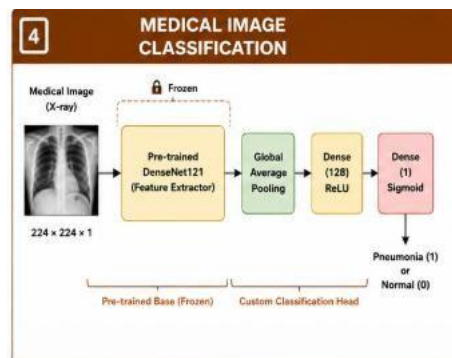
Problem

Design and develop a transfer learning-based model for classifying medical images such as pneumonia, skin disease or brain tumor images. Evaluate accuracy and other performance metrics.

Answer

A transfer learning model can be used to classify **chest X-ray images for pneumonia detection**. A pre-trained **DenseNet121** can be used because CNNs are effective at extracting visual patterns from medical images.

Architecture



Data Preprocessing

1. Resize X-ray images.
2. Normalize pixel values.
3. Remove corrupted images.
4. Split data into training, validation and testing sets.
5. Apply augmentation such as rotation and zoom to the training data. **Model**

Training

The pre-trained DenseNet layers are initially frozen.

DenseNet121 → Frozen



Custom Layers → Trainable



Prediction

After initial training, selected deeper layers can be unfrozen for fine-tuning. **Output**

For binary classification:

Output = Sigmoid

Probability > 0.5 → Pneumonia

Probability ≤ 0.5 → Normal

The threshold can be adjusted depending on the required medical sensitivity.

Evaluation Metrics

Accuracy alone is not sufficient for medical diagnosis.

Important metrics are:

Precision

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall/Sensitivity

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-score

$$\text{F1} = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Also use:

- Confusion Matrix
- Specificity
- ROC-AUC

Importance of Recall

For disease detection, high recall is important because missing an actual disease case can be more serious than generating a false positive.

Conclusion

Transfer learning with DenseNet can provide an efficient medical image classification system. Evaluation should focus not only on accuracy but also on recall, precision, F1-score and other clinically relevant metrics.

QUESTION 5 – RNN FOR TEXT CLASSIFICATION

Problem

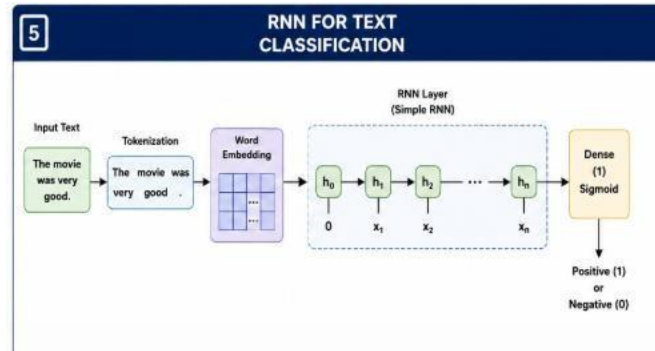
Design and develop a Recurrent Neural Network (RNN) model for text classification and explain how sequential information is processed.

Answer

An RNN is designed to process sequential data such as text. It maintains a hidden state that carries information from previous words.

For example, an RNN can classify movie reviews as **positive or negative**.

Architecture



Example

Sentence:

"The movie was very good."

The RNN processes the words sequentially:

The → movie → was → very → good

↓ ↓ ↓ ↓ ↓ h_1
 h_2 h_3 h_4 h_5

Each hidden state depends on the current word and the previous hidden state. **RNN**

Formula

$$h_t = \tanh (W_x x_t + W_h h_{t-1} + b)$$

where:

- (x_t) = current input
- (h_{t-1}) = previous hidden state
- (h_t) = current hidden state
- (W_x) = input weight
- (W_h) = recurrent weight
- (b) = bias

Training Steps

1. Collect text dataset.
2. Clean the text.
3. Tokenize sentences.
4. Convert words into numerical sequences.
5. Apply padding.
6. Use word embeddings.
7. Pass sequences through the RNN.

8. Add a classification layer.
9. Train using backpropagation through time.
10. Evaluate the model. **Example Code**

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, SimpleRNN, Dense
```

```
model = Sequential([
    Embedding(10000, 64),
    SimpleRNN(64),
    Dense(1, activation="sigmoid")
])
```

```
model.compile(optimizer="adam",
              loss="binary_crossentropy",
              metrics=["accuracy"])
```

```
model.summary()
```

Evaluation Use:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Advantages of RNN

- Handles sequential data.
- Maintains information from previous time steps.
- Useful for text classification.
- Can process variable-length sequences.

Limitation

Traditional RNNs may struggle with long-term dependencies because of the **vanishing gradient problem**. LSTM or GRU networks can be used when longer contextual information is required.

Conclusion

RNNs are suitable for text classification because they process words sequentially and maintain information from previous words through their hidden states.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____